

H-4.2: Asymmetric deceleration maneuvers aircraft toward other objects	SC-6.2: Asymmetric deceleration must not lead to loss of directional control or cause aircraft to depart taxiway, runway, or apron
H-4.3: Deceleration occurs after V1 point during takeoff	SC-6.3: Deceleration must not be provided after V1 point during takeoff

Modeling the control structure

The next step in STPA is to model the hierarchical control structure, as Figure 2.5 shows.

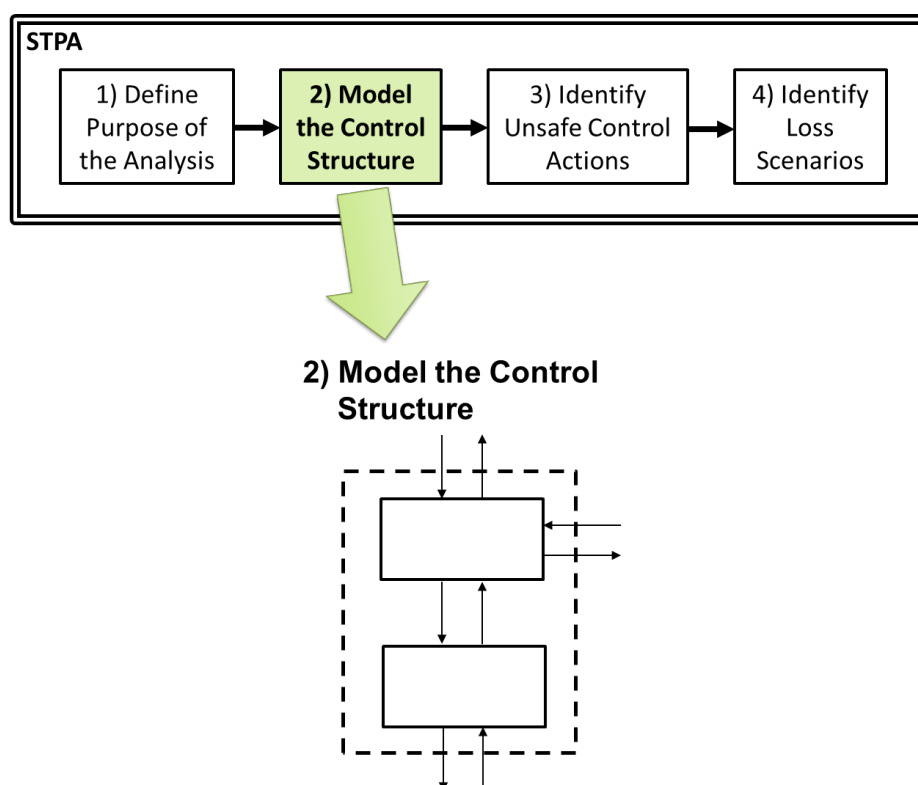


Figure 2.5: Modeling the control structure

What is a control structure?

Definition: A hierarchical control structure is a system model that is composed of feedback control loops. An effective control structure will enforce constraints on the behavior of the overall system.

A hierarchical control structure is composed of control loops like the one shown in Figure 2.6. In general, a controller makes decisions to achieve goals and provides control actions to control some process and to enforce constraints on the behavior of the controlled process. The controlled process is any process that is controlled, such as a physical process or another controller. The control algorithm represents the controller's decision-making process—it determines the control actions to provide.

Controllers also have process models that represent the controller's internal beliefs used to make decisions. Process models may include beliefs about the process being controlled or other relevant aspects of the system or the environment. Process models may be updated in part by feedback used to observe the controlled process.

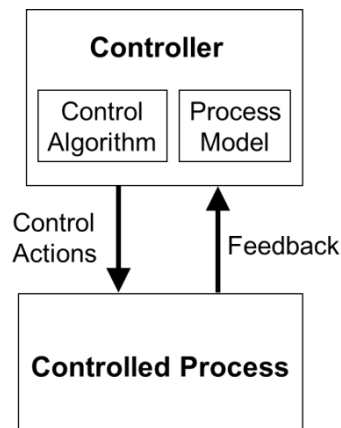


Figure 2.6: Generic control loop

Problems can occur at any point in Figure 2.6. For example, a process model that is not consistent with reality (e.g. a controller believes an aircraft is descending when it is really ascending or believes a car is in Park when it is really in Reverse, the airport runway is empty, etc.) can lead to control actions that are unsafe. A sensor failure may cause incorrect feedback and lead to unsafe behavior. A design may be missing the necessary feedback or may provide delayed feedback that results in a process model flaw and unsafe behavior. As you'll see, STPA provides a way to systematically identify these and other scenarios that can lead to a loss.

The generic control loop in Figure 2.6 can be used to explain and anticipate complex software and human interactions that can lead to losses—two of the biggest challenges in modern engineering. For humans, the process model is usually called a mental model and the control algorithms may be called operating procedures or decision-making rules, but the basic concept is the same.

Of course, most systems typically have several overlapping and interacting control loops. Multiple interacting control loops can be modeled in a hierarchical control structure, as shown in Figure 2.7. A simple example for aviation is shown in Figure 2.8.

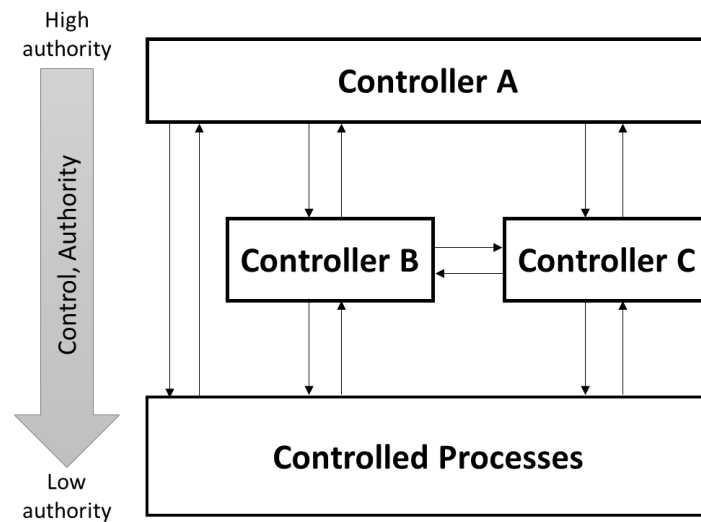


Figure 2.7: Generic hierarchical control structure

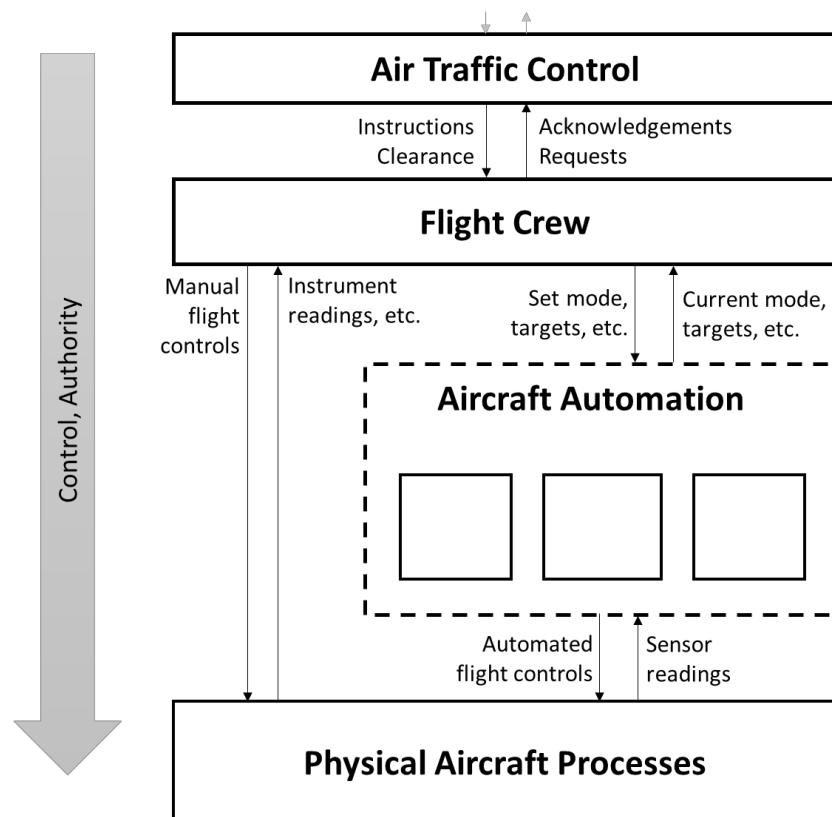


Figure 2.8: Simple example of a hierarchical control structure for aviation

In general, a hierarchical control structure contains at least five types of elements⁷:

- Controllers
- Control Actions
- Feedback
- Other inputs to and outputs from components (neither control nor feedback)
- Controlled processes

The vertical axis in a hierarchical control structure is meaningful: it indicates control and authority within the system. The vertical placement represents the hierarchy of control from high-level controllers at the top to the lowest-level entities at the bottom. Each entity has control and authority over the entities immediately below it, and each entity is likewise subject to control and authority from the entities immediately above. For example, the aircraft automation in Figure 2.8 can act as a controller by sending control actions to physical aircraft systems and monitoring feedback. At the same time, the aircraft automation is also a controlled process that receives and executes control actions from the flight crew and sends feedback to the crew.

In other words, all downward arrows represent control actions (commands) while the upward arrows represent feedback. These conventions help to manage complexity and make control relationships and feedback loops easier to recognize.⁸ In some cases, simply drawing a control structure diagram can make previously undiscovered flaws painfully obvious. For example, control actions may be provided by entities that don't have the necessary feedback to select safe control actions, feedback may be provided to entities with no ability to do anything about it, multiple controllers may be able to provide conflicting commands to the same component with no ability to detect or resolve the conflict, etc. If you do not catch flaws at this stage though, don't worry—the rest of the STPA method will systematically identify these and other issues.

Common Points of Confusion

A control structure is not a physical model

The hierarchical control structure used in STPA is a functional model, not a physical model like a physical block diagram, a schematic, or a piping and instrumentation diagram. The connections show information that can be sent, such as commands and feedback—they do not necessarily correspond to physical connections. For example, the interactions between the flight crew and air traffic control are not of a physical nature, but they are modeled in a functional control structure.

A control structure is not an executable model

The control structure is not an executable model or a simulation model. In fact, control structures often include components for which executable models do not exist (such as humans). Instead, STPA can be used to carefully derive the necessary behavioral constraints, requirements, and specifications needed to enforce the desired system properties. For example, the control structure in Figure 2.8 does not assume that air traffic control will always send instructions to the flight crew when needed, that they will always have the capability to send instructions (e.g. that radios will always be operational), that the correct instructions will always be sent, or that the instructions will always be followed by the pilots. It simply indicates that a system was/will be created to allow air traffic control to send instructions to the

⁷ A control structure can also contain actuators and sensors, but these are usually added later during the scenario identification step

⁸ Control structures can also be drawn left to right if needed, but we are purposely simplifying the discussion here.

flight crew. The next steps in STPA will carefully examine how unsafe behaviors may occur, including instructions that are sent but not received, unsafe instructions that may be sent, etc. Although the control structure itself is not an executable model, the STPA method will produce precise requirements and other outputs that can be used to generate executable models and specifications.

A control structure does not assume obedience

Do not confuse controllers and control actions with obedience. Just because a controller sends a control action does not mean that in practice it will always be followed. Likewise, just because a feedback path is included in a control structure does not mean that in practice the feedback will always be sent when needed or that it will be accurate. The control actions and feedback in a control structure simply indicate that a mechanism will be created to send this information (that is, it will be in the system design). It does not imply or assume anything about how controllers and processes will actually behave in practice. In fact, a major goal of STPA is to analyze the control structure and anticipate how each element might behave in unsafe and potentially unexpected ways.

Use abstraction to manage complexity

One of the biggest challenges in any hazard analysis is managing system complexity. Control structures use abstraction in several ways to help manage complexity. For example, in commercial flights the flight crew might consist of two or three individual pilots. Rather than clutter the control structure right from the beginning with three separate pilots, we can group them together as a flight crew that collectively provides control actions and collects feedback. Similarly, rather than explicitly listing every individual aircraft subsystem, we could begin at a more abstract level by modeling aircraft automation and the physical processes they control as two levels in the control hierarchy.

The principle of abstraction can also be applied to the command and feedback paths in the control structure. Rather than listing each individual button, switch, and lever in the cockpit, we might begin with much broader actions like a climb maneuver. Later, we could refine these broad actions into pitch, thrust, or other commands as appropriate. This principle is especially useful during early development phases when individual commands and sensors are not yet known.

The control action path may contain mechanisms by which the controller acts upon a controlled process (referred to as actuators) and mechanisms by which the controller senses feedback from a controlled process (referred to as sensors). These details are usually abstracted away when initially creating the control structure, but the control structure will be refined to include actuators and sensors later during the scenario creation step.

One additional type of abstraction is used in control structures. Consider the flight commands sent from the flight crew to the aircraft automation. In a remotely piloted UAV application, those commands may need to pass through many different components—from a physical button on the command console, through an embedded system that encodes the command within a digital packet, to a network switch, a radio transmitter, a satellite, and a radio receiver on the UAV. It is not necessary to show all of these detailed steps along the control path in the initial control structure—what matters is that the remote pilot will have some way to send flight commands to the UAV.

In fact, the most efficient way to apply STPA is to begin before those design decisions have been made and before such details are known. The abstract control structure above can be used to begin STPA and identify the requirements and constraints for the communication path and other parts of the system. Then, STPA results can be used to drive the architecture, preliminary and detailed design, make implementation decisions, and refine the control structure. Even if details are known and design decisions have been made, it can be helpful to first apply STPA at a higher abstract level first to provide quicker results and identify broader issues before analyzing more detailed control structure models.

Modeling the control structure

Control structure modeling begins with an abstract control structure and iteratively adds detail. In many cases, the control structure and the control loops within the system may be obvious or can be reused from previous applications. If the control structure is not obvious, the following guidance explains one way to model a functional control structure and add detail. Appendix B contains examples of control structures that have been used in different industries.

One way to begin is to identify basic subsystems needed to enforce the constraints and prevent the hazards identified earlier. For example, we derived hazards and constraints related to insufficient deceleration. These constraints could be enforced by using a wheel braking subsystem, reverse thrust, and other subsystems. [Figure 2.9](#) shows an initial control structure with these subsystems.

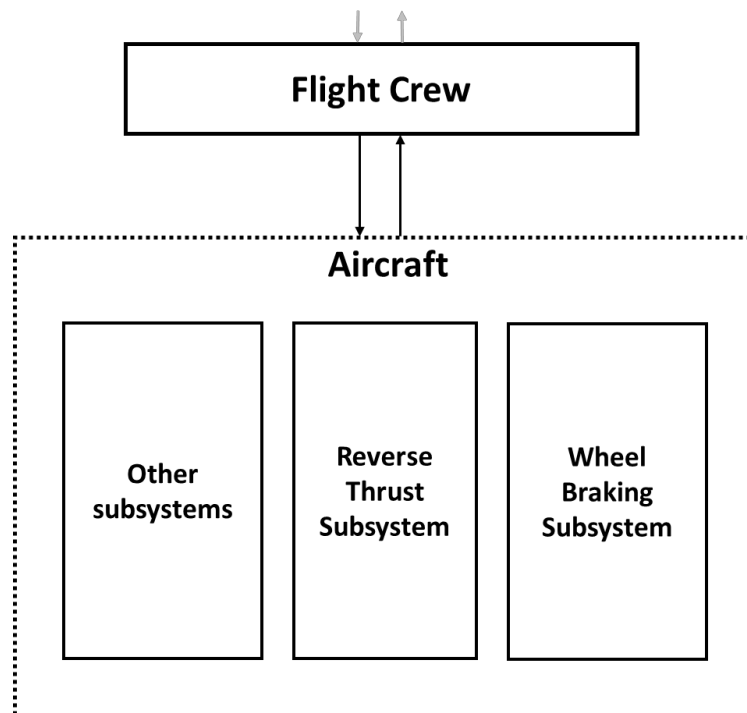


Figure 2.9: Example of a refined control structure with subsystems

The control structure can be refined by defining how each subsystem will be controlled. For the purposes of this example, the wheel braking subsystem will be refined. Will the wheel braking subsystem be controlled directly and manually by the flight crew only? Will control units, automated controllers, or other humans exist that can also control the brakes? If STPA is being applied during later development stages, these answers may be known and obvious. If STPA is being applied during early concept development, then the hazards and the constraints above can be used to guide these decisions. For example, we may include an automated brake controller in the wheel braking subsystem to automatically brake upon landing or rejected takeoff.

Figure 2.10 shows the refined control structure including a Brake System Control Unit (BSCU) within the wheel braking subsystem. Notice that we are carefully “zooming in” to add more detail in the control structure.

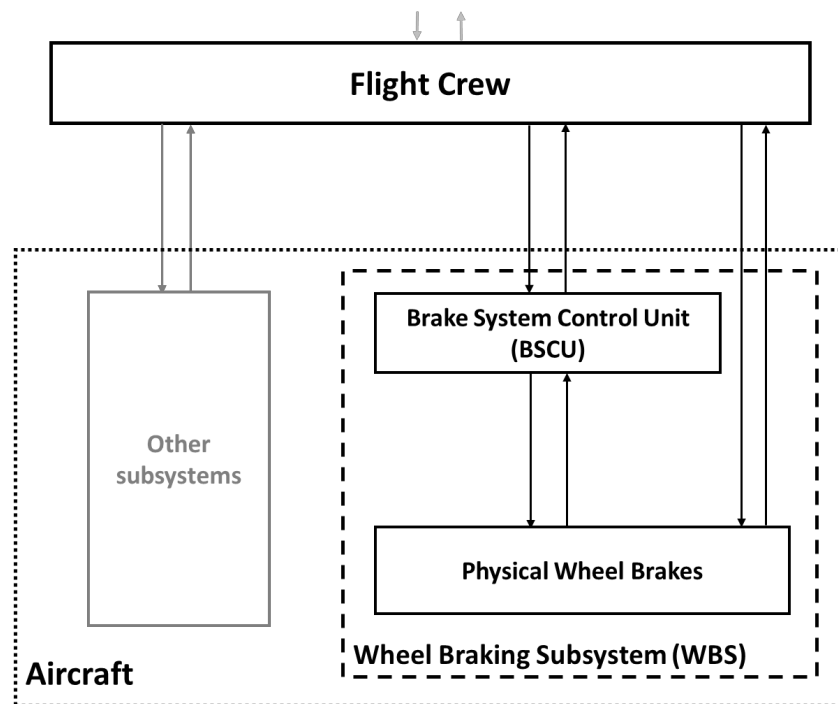


Figure 2.10: Refined control structure with subsystem controllers

During control structure development, responsibilities can be assigned to each control structure entity. These responsibilities are a refinement of the system-level constraints—what does each entity need to do so that *together* the system-level constraints will be enforced? For example, the BSCU might be responsible for automatically pulsing the brakes when braking causes a skid (anti-skid function) while the flight crew may be responsible for deciding when the brakes need to be applied.

Example responsibilities related to wheel braking:

Physical Wheel Brakes

- R-1: Decelerate wheels when commanded by BSCU or Flight Crew [SC-6.1]

BSCU

- R-2: Actuate brakes when requested by flight crew [SC-6.1]
- R-3: Pulse brakes in case of a skid (Anti-skid) [SC-6.2]
- R-4: Automatically engage brakes on landing or rejected takeoff (Autobrake) [SC-6.1]

Flight crew

- R-5: Decide when braking is needed [SC-6.1, SC-6.3]
- R-6: Decide how braking will be done: Autobrake, normal braking, or manual braking [SC-6.1]
- R-7: Configure BSCU and Autobrake to prepare for braking [SC-6.1]
- R-8: Monitor braking and disable BSCU, manually brake in case of malfunction [SC-6.1, SC-6.2]

Control actions for each controller can be defined based on these responsibilities. For example, the flight crew will need the capability to send manual braking control actions to satisfy R-5 and R-6. They will need a way to arm and set the BSCU to satisfy R-6 and R-7. They may need to disarm the BSCU to

satisfy R-8. Figure 2.11 shows a revised control structure with labeled control actions based on the responsibilities.

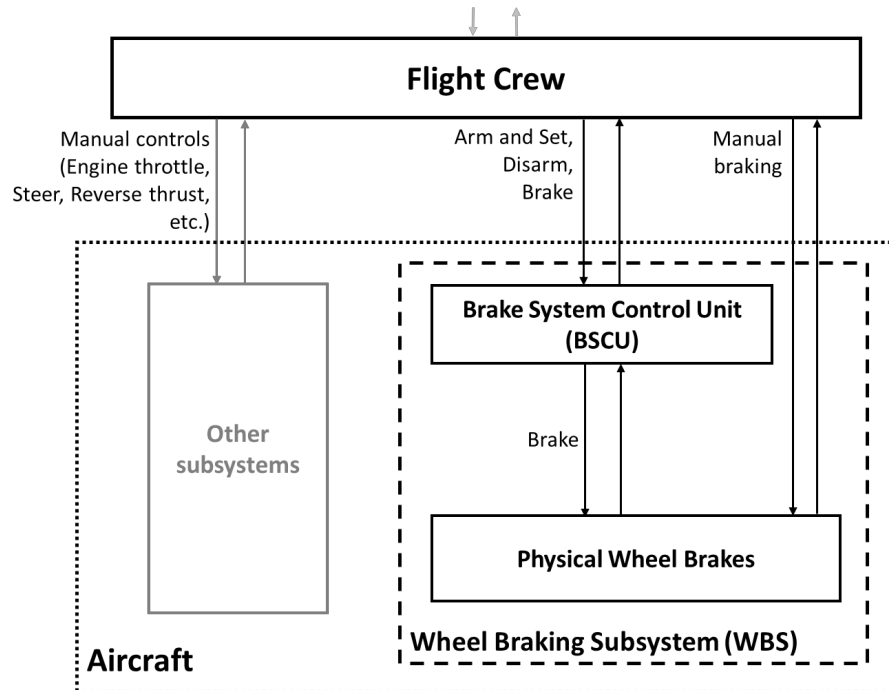


Figure 2.11: Refined control structure after allocation of processes to subsystems

At this point, the controllers and control actions have been identified and labeled but what about feedback? Feedback can be derived from the control actions and responsibilities by identifying the process models that controllers will need to make decisions. Then, feedback and other information needed to form accurate process models can be identified. For example, R-3 specifies that the BSCU will need to pulse the brakes in case of a skid. To do this, the BSCU will need to know that a skid is occurring (information that should be included in the BSCU's process model). What feedback is needed to detect a skid? Wheel speed feedback could be used. Similarly, R-8 specifies that the flight crew may need to disable the BSCU in case of a malfunction. To do this, the flight crew will need to know that the BSCU is malfunctioning (again, information that should be in the process model). What feedback can the flight crew use to detect a malfunction? Feedback about BSCU faults could be provided. To automatically engage the brakes upon landing or RTO (R-4), the BSCU will need to know when the aircraft has landed or when RTO is occurring (should be in the process model). Weight on wheels switches and other inputs could be used to detect landing and RTO conditions. Table 2.2 below shows how feedback can be derived from the responsibilities.

Table 2.2: Examples showing how feedback can be derived from responsibilities

BSCU Responsibility	Process Model	Feedback
Actuate brakes when requested by flight crew [SC-6.1]	Braking is requested by flight crew	Brake pedal applied
Pulse brakes in case of a skid (Anti-skid) [SC-6.2]	Aircraft is skidding	Wheel speeds Inertial reference unit
Automatically engage brakes on landing or RTO (Autobrake) [SC-6.1]	Aircraft landed Takeoff is rejected	Weight on wheels Throttle lever angle

The control structure can be refined further by using the responsibilities to “zoom in” again and add additional details. For example, the physical wheel brakes are responsible for decelerating the wheels on command (R-1). This could be done with hydraulics. R-6 indicates that the BSCU will need to execute both normal and automatic braking (Autobrake). Two controllers within the BSCU could be used to control these two behaviors: an autobrake controller and a hydraulic controller. Two controllers within the BSCU could be used to control these two behaviors: an autobrake controller and a hydraulic controller.

Figure 2.12 shows a refined control structure with feedback labeled and with BSCU internal controllers identified.

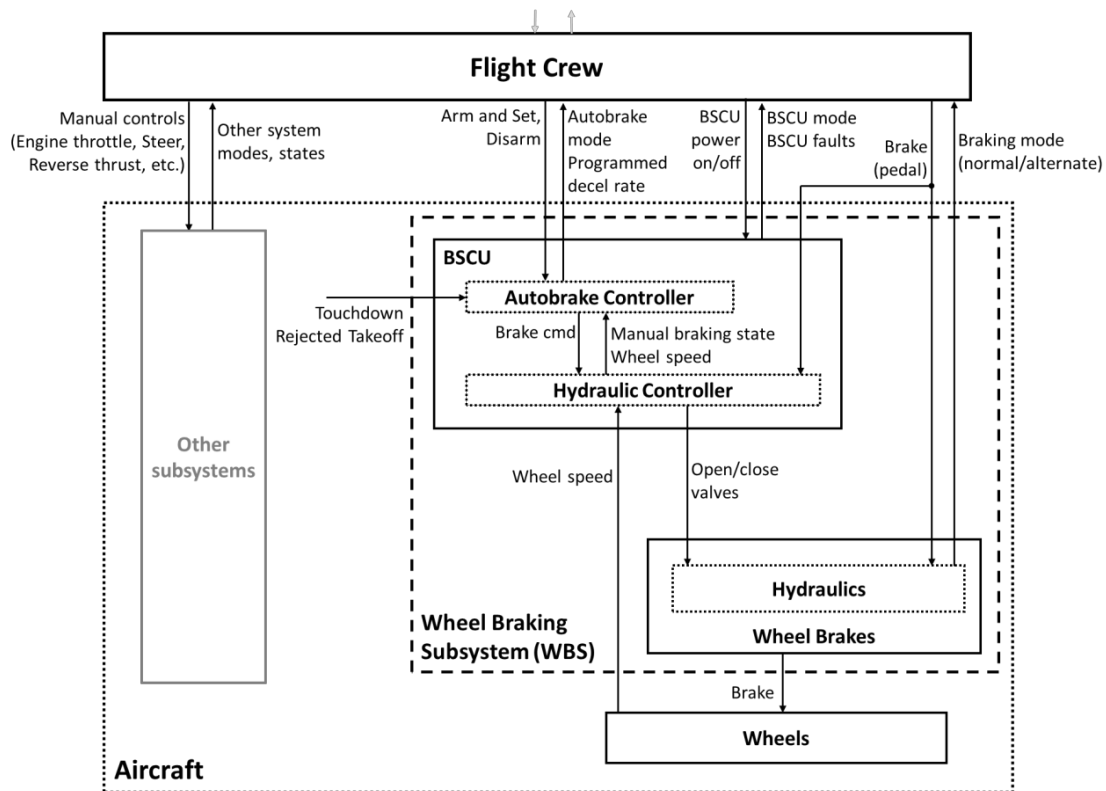


Figure 2.12: Control structure after refinement based on the responsibilities

Common questions when modeling the control structure

Does the control structure need to be complete before proceeding?

If STPA is being applied early before development is finished, some information may not be known and the control structure may be incomplete. The analysis can begin with an incomplete control structure and STPA will help identify potentially missing feedback, controls, and other gaps so the control structure can be refined in parallel with development. The bare minimum needed to begin the next steps is at least one controller, control action, and controlled process. However, in general STPA will be easier and more efficient if relevant information is not intentionally missing from the control structure.

If STPA is being applied during later development stages after decisions about controllers, control actions, and feedback have already been made, then there is no reason to intentionally omit high-level information from the control structure. STPA can be applied to the design as it exists to identify potential flaws like critical feedback that was overlooked.

How specific should the arrow labels be?

Avoid using ambiguous and vague labels in the control structure, such as labeling all control actions as simply “Commands”, all feedback as simply “Feedback” or “Status”, and controllers as simply “Computer” or “Controller.” Labels for control actions should indicate the type of command (if known), like “Open/close valves” and feedback should indicate the type of information being sent (if known), like “Wheel speed”. The specific physical medium used to send control actions and feedback is irrelevant at this point—what matters is the *functional* information that can be sent. For example, use “BSCU power on/off” rather than “BSCU button”, “Shared bus”, or “Encoded digital packets.” Similarly, labels for controllers should indicate the functional type of behavior or the role of the controller, not the physical implementation. For example, use controller labels like “Autobrake Controller” rather than “Single Board Computer,” and use labels like “Flight Crew” or “Pilot” whether the piloting function is accomplished on board or on the ground.

Should the control structure include all actuators and sensors?

Specific actuators and sensors are not needed to begin STPA and do not need to be included yet (they will be included in a later step). To help manage complexity, it can be helpful to wait until these details need to be considered during the scenario creation step later in STPA. For example, the wheel braking subsystem control actions and feedback in Figure 2.12 may be implemented with different kinds of electromechanical valves or by using some other completely different implementation. What matters at this phase are the types of commands and feedback that might be provided, not the specific implementation (which may or may not be known).

How do physical processes and physical interactions fit into the control structure?

Like any model, a control structure model emphasizes certain aspects of the real world while abstracting or de-emphasizing others. A control structure will emphasize functional relationships and functional interactions, which is very useful for identifying problems like design flaws, requirements flaws, human error, software errors, and even traditional physical component failures. A control structure model does not typically capture purely physical or geometric relationships like physical proximity between components or fire propagation.

The physical processes being controlled are typically specified at the lowest level of the control structure while every level above specifies functional controllers that make decisions and directly or indirectly control the physical processes.

Does the control structure require a linear hierarchy?

No. For some systems, the control structure may appear similar to a ladder with a clear vertical linear hierarchy. Control structures for other systems may include multiple controllers at the same vertical level that do not control each other. Controllers may all control the same physical process, or they may control different processes.

Controllers at the same level may also communicate with each other outside of a control/feedback relationship. Such communication may be represented by horizontal arrows in the control structure diagram. In general, the interactions (arrows) in a control structure can represent any of three categories: Control actions, feedback, and other information.

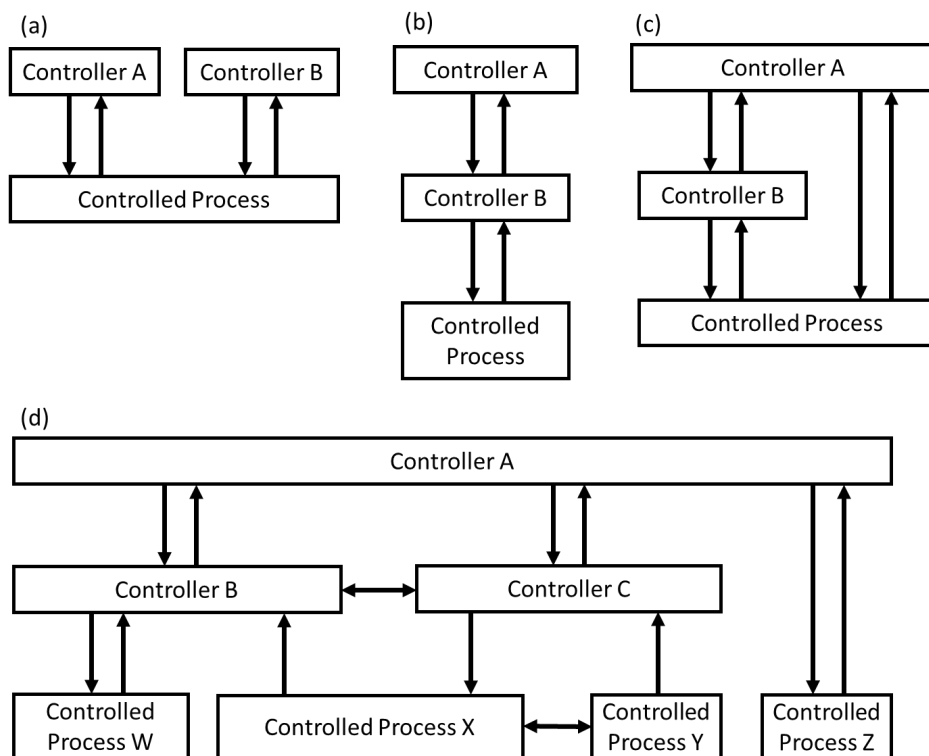


Figure 2.13: Different types of control structures

Figure 2.13 shows a few different types of control structure models, each with different advantages and tradeoffs to consider when designing or analyzing a system. The control structure in (a) shows two controllers operating in parallel to control and monitor a shared process. The controllers do not interact with each other and neither one can control the other (which may lead to loss scenarios that STPA will identify). This type of structure could be advantageous if controllers will need to act quickly or independently of each other—for example, it might be advantageous to provide several operators with the capability to directly abort an operation whenever an unsafe condition is observed without going through a chain of command. However, this structure can present coordination challenges like diffusion of responsibility and assumptions about the actions of other controllers (later steps will analyze these and other problems in more detail).

The control structure in (b) shows a linear control hierarchy such as a clear chain of command. This structure may help address the coordination issue and clarify the difference in responsibilities for each

controller, but may present challenges in terms of reaction time for Controller A's feedback and control actions and is more dependent on the operation of Controller B.

Control actions and feedback may "skip" over levels—there is no model limitation that prevents controllers from interacting beyond one level above and below. For example, high-level controllers may be able to bypass other controllers and directly control or monitor lower-level processes as in (c). Controller A could represent pilots that normally use automation in Controller B to actuate the brakes. However, in an emergency the pilots may be able to bypass Controller B and directly (manually) actuate the brakes.

More complex structures like (d) are also possible. There is no requirement for a 1-to-1 mapping between controllers and processes. A controller may control one or more processes, and a process may be controlled by zero or more controllers. Typically, each control action path will be paired with a parallel feedback path but not always. For example, Controller C may not be able to control Process Y directly. Instead, it may control Process X and monitor the effect of X on Process Y.

How do I figure out who controls who?

In many cases the control relationship is simple and obvious, such as a supervisor giving instructions to a subordinate or a computer sending commands to open/close a valve. Control relationships can also be less direct, such as managers assigning priorities, airlines providing standard operating procedures, etc. These are all different forms of control.

As mentioned earlier, control is different from obedience. Just because a control relationship exists does not mean that the control actions will always be executed and performed adequately. In fact, we may not even want all control actions to be obeyed all the time; control actions may potentially be disregarded for good reasons when required to prevent a loss in some unexpected situation. STPA does not assume obedience and these problems will be carefully examined in later steps.

Similarly, the ability to trigger a response or influence behavior of another component does not automatically imply control. A control action to open a valve may trigger the valve to open, but conversely a feedback signal indicating high temperature may trigger a controller to turn on a fan. The ability to influence or respond is not sufficient to distinguish control actions from feedback.

Control involves making purposeful decisions to achieve goals. Feedback can be provided without being aware of (or responsible for) a particular higher-level goal, but control actions are always provided to achieve a goal. Control also typically involves supervision—monitoring lower-level entities, having better capability or information to evaluate their behaviors or impacts, and guiding them or intervening to ensure higher-level goals are achieved.

The control hierarchy is closely related to the hierarchy of goals and responsibilities. The automated BSCU controller above may have very narrow responsibilities related to triggering brakes when a landing is detected. The flight crew has higher-level responsibilities like deciding when to land and higher-level goals like avoiding a rough landing. The BSCU may be a necessary but insufficient element in achieving these higher-level flight crew goals—the flight crew must supervise the BSCU and many other elements to achieve the higher-level goals. Similarly, air traffic control has even higher-level responsibilities like ensuring minimum separation between multiple aircraft and higher-level goals like maximizing throughput across many flights. A single flight crew is only one element used by air traffic control to achieve the goals, and the flight crew might not even be aware of how they are contributing to those higher-level goals or if those higher-level goals are being achieved by air traffic control.

Finally, control is closely related to authority. Consider the relationship between air traffic control and pilots. Air traffic control has authority over pilots and provides instructions and clearances that pilots generally must obey. Pilots do not have authority over air traffic control, and they cannot simply order air traffic control to obey. Pilots *can* declare an emergency and air traffic control will need to respond appropriately to that feedback—just because pilots can trigger a response does not imply

control. Pilots can also disobey air traffic control instructions when necessary to ensure safety of the flight, but obedience is also different from control. Pilots do have final and direct authority over their aircraft, but they do not have final authority over ATC and they do not manage the overall airspace. ATC generally controls aircraft indirectly by issuing instructions and clearances, not the other way around. Chapter 5 contains further discussion about control structures and control relationships as they relate to organizational and social analysis.

Although beginners sometimes struggle with these distinctions, with more experience it quickly becomes easy and natural. It can also be helpful to realize that mistakes about who controls whom in the control structure usually do not have a significant impact on the results of the analysis. For example, suppose control action X is mischaracterized as feedback X. Because it is characterized as feedback, the step that identifies unsafe control actions will not consider how a missing or delayed control action X might lead to a hazard. However, the next step examines potential feedback problems and will identify the same scenarios when considering how missing or delayed feedback X might lead to a hazard.

Do I need to document anything other than the control structure diagram?

It is good practice to document any additional information that is helpful to understand the control structure. Additional information about controllers such as a basic description, purpose, special functionality, controller responsibilities, process models, etc. should be documented for a completely specified control structure. Clarifying information about control actions and feedback can also be helpful, particularly when important aspects may not be clear from short labels. The same is true for controlled processes. In general, any important clarifying information or assumptions should be clearly documented along with the control structure diagram.

What should I look for when reviewing a control structure?

The following tips can help find common mistakes in a control structure:

Tips to prevent common mistakes in a control structure

- Ensure labels describe functional information that is sent, not a specific physical implementation.
- Avoid ambiguous and vague labels like simply "Command" or "Feedback" when the type of information is known.
- Check that every controlled physical process is controlled by one or more controllers (not always required, but often indicates a mistake).
- Review responsibilities (including traceability) for conflicts and gaps.
- Check that control actions needed to satisfy the responsibilities are included.
- Check that feedback needed to satisfy the responsibilities is included. (optional if applied early in concept development when feedback is unknown; later steps can identify missing feedback)

Identifying Unsafe Control Actions

Once the control structure has been modeled, the next step (shown in Figure 2.14) is to identify Unsafe Control Actions.